

Fig. 8: Author's prototype showing attachment of the IR module to a bicycle wheel for speed measurement

in the program based on the number of fan blades. Most ceiling fans have three blades. The program contains alternate lines in the code for three-blade and four-blade fans.

### Bicycle tachometer

The speed of a bicycle can be measured by counting the number of wheel rotations. Each rotation moves the bicycle forward by a distance equal to  $\pi \times D$ , where  $D$  is the wheel diameter.

If the number of rotations is counted for four seconds, the rotations per minute become  $15 \times n$ , where  $n$  is the number of rotations counted in four seconds.

The distance travelled in one minute is therefore  $3.14 \times D \times 15 \times n$ .

For a typical bicycle, the rear wheel diameter is about 24-25 inches ( $\approx 62.5\text{cm}$ ). Substituting this value, the speed works out to approximately:

$$\text{Speed} \approx 1.76 \times n \text{ km/hr}$$

Thus, the bicycle speed can be obtained directly from the number of rotations.

The module is attached to the front wheel using a clamp on the same side as the brake shoe. The projection of the air-pump nozzle is fitted with an L-clamp to bring it closer to the module than the spokes.

The module is fixed temporarily with tape so that adjustments can be made to align it properly with the metal face of the L-clamp within the IR sensing range.

In this way, the wheel rotation is sensed only once per turn. The spokes should not be counted. This can be ensured by adjusting the module's preset so that it responds only to the reflection from the L-clamp face.

The wires from the module are connected to the PCB that contains the battery and the TM1637 display board.

**EFT note.** The program can be modified using the above formula to measure vehicle speed for different wheel sizes by changing the wheel diameter value in the code. **EFY**

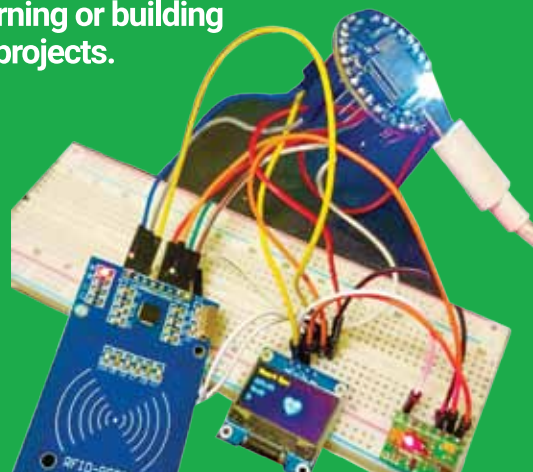


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